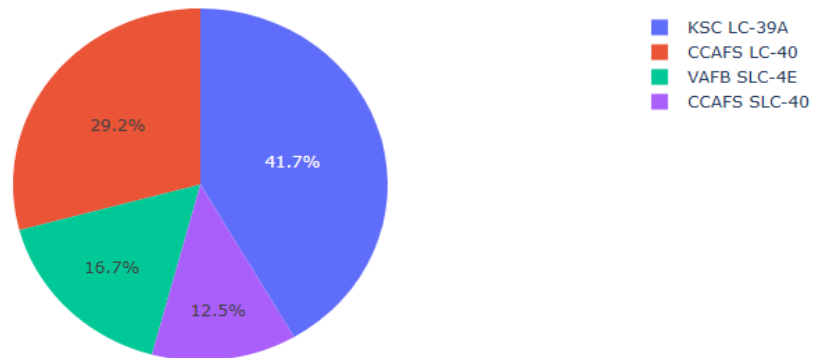


All Sites

×

Total Success By Site



SpaceX Falcon 9 Landing Prediction

31.12.2025

—

Syed Badar

Executive Summary

This project aims to predict the success of SpaceX Falcon 9 first-stage landings. By leveraging the SpaceX API and web scraping, we analyzed launch data to build machine learning models. Our final models achieved an **83.33% accuracy**, providing a reliable tool for estimating launch costs based on booster recovery.

Methodology

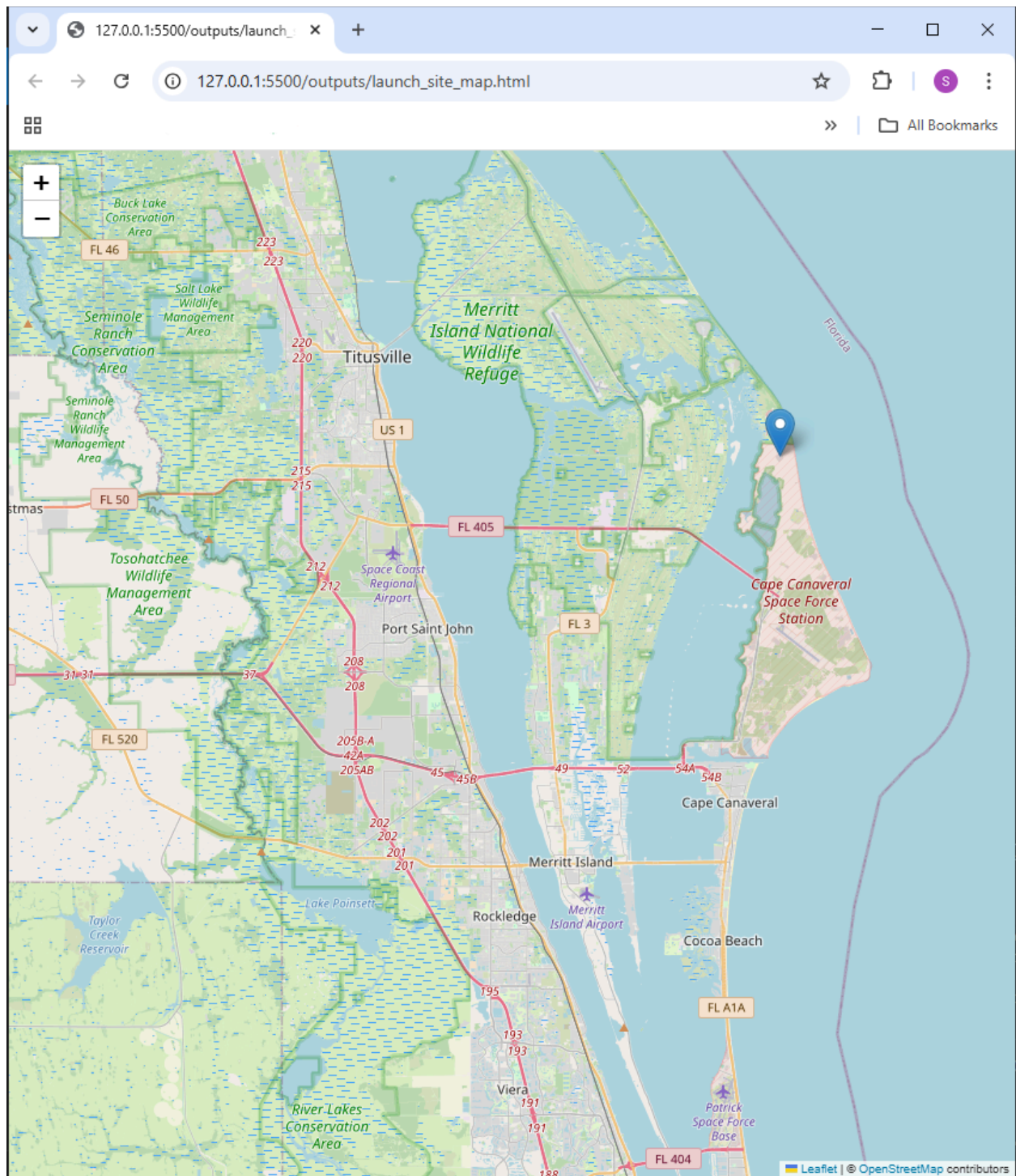
1. **Data Collection:** Used SpaceX API and BeautifulSoup for Wikipedia scraping.
2. **EDA:** Conducted analysis using SQL (SQLite) and visualization (Seaborn/Matplotlib).
3. **Interactive Analytics:** Built geospatial maps with **Folium** and a dashboard with **Plotly Dash**.
4. **Predictive Analysis:** Built and tuned Logistic Regression, SVM, Decision Tree, and KNN models using GridSearchCV.

Exploratory Data Analysis (EDA)

1. **Data Collection:** Used SpaceX API and BeautifulSoup for Wikipedia scraping.
2. **EDA:** Conducted analysis using SQL (SQLite) and visualization (Seaborn/Matplotlib).
3. **Interactive Analytics:** Built geospatial maps with **Folium** and a dashboard with **Plotly Dash**.
4. **Predictive Analysis:** Built and tuned Logistic Regression, SVM, Decision Tree, and KNN models using GridSearchCV.

Interactive Visual Analytics

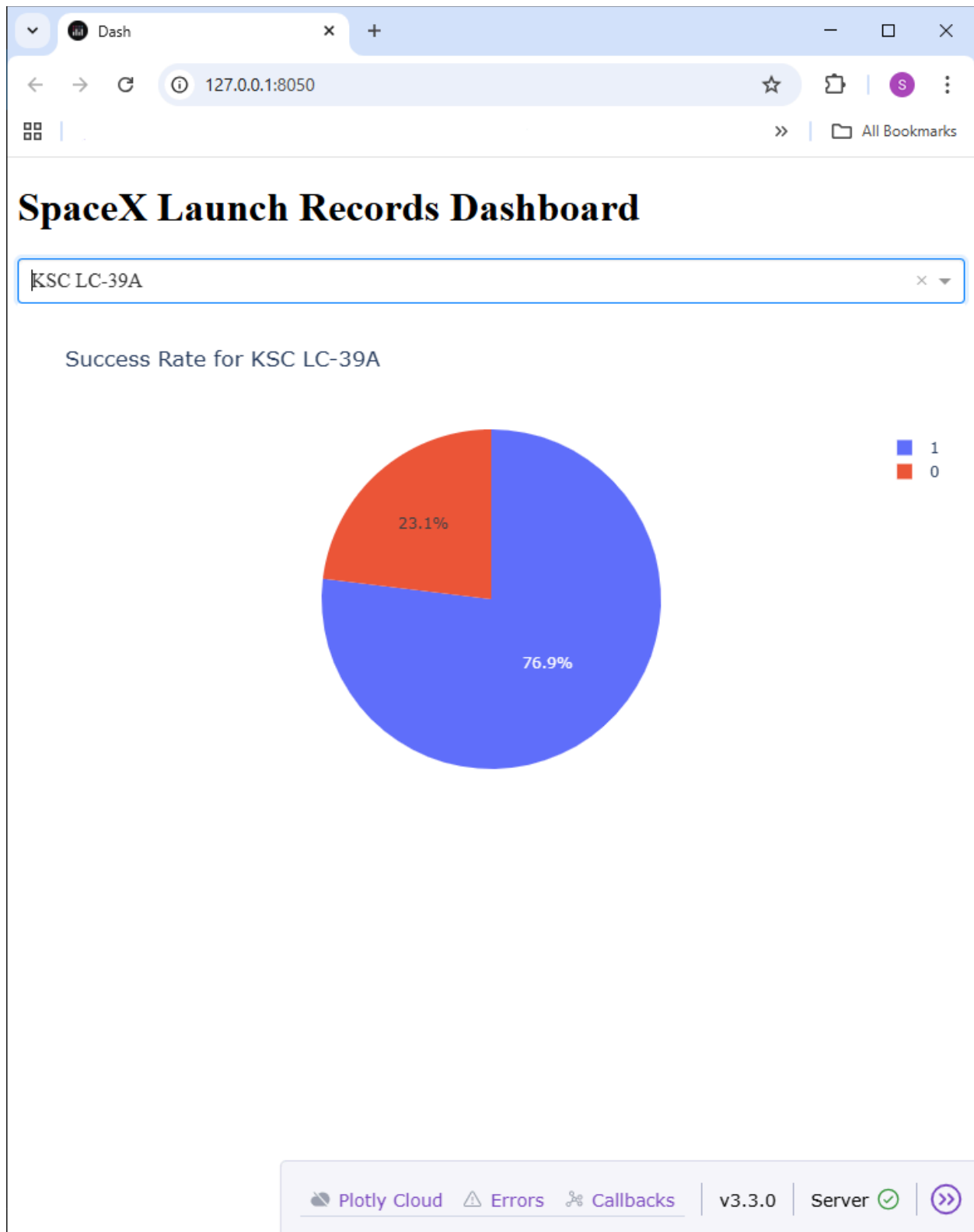
1. **Data Collection:** Used SpaceX API and BeautifulSoup for Wikipedia scraping.
2. **Dash Dashboard:** An interactive application allowing users to filter success rates by launch site and payload mass.

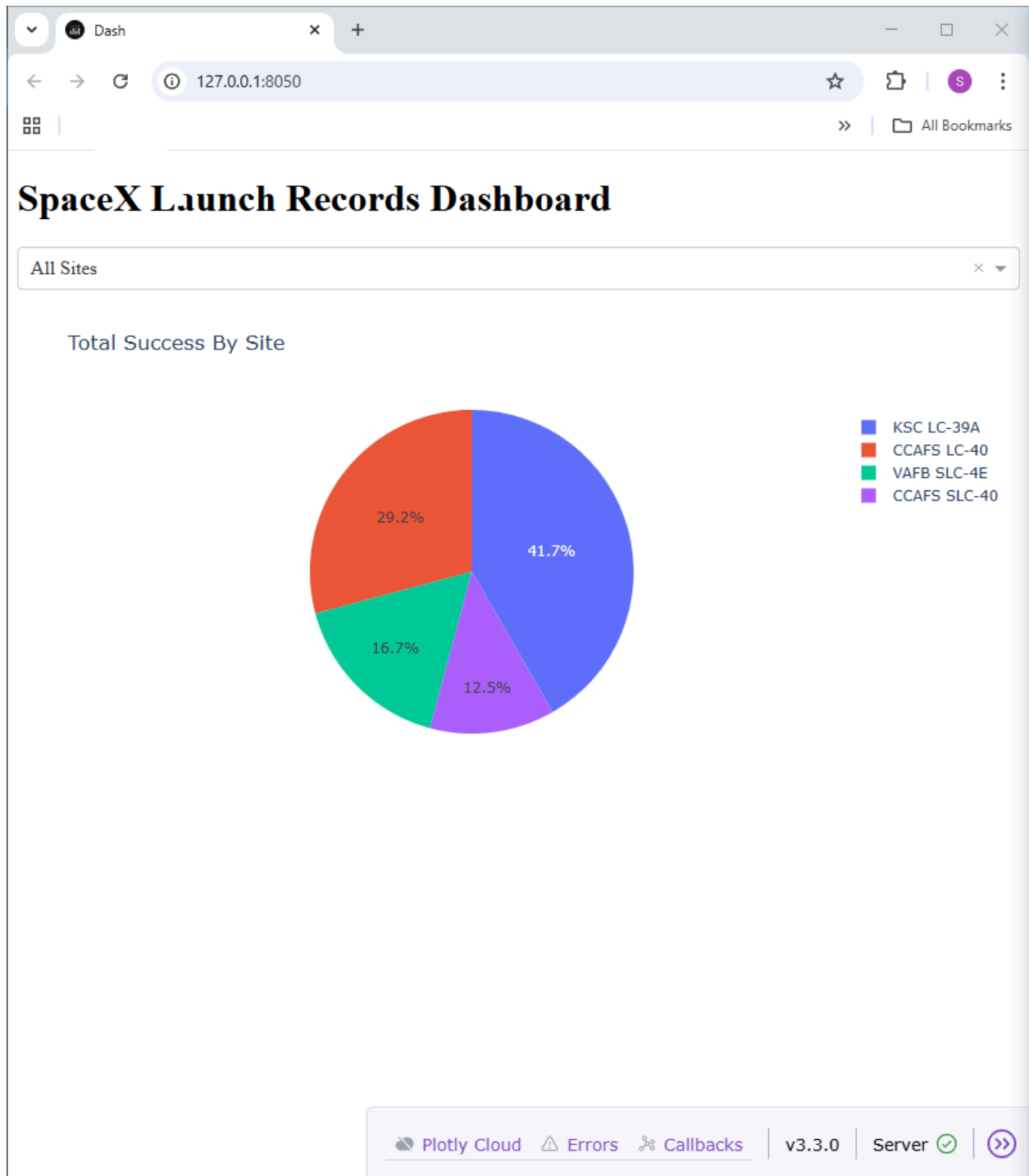


Machine Learning Results

We compared four classification models. All models converged on the same test accuracy, indicating robust feature selection.

Model	Test Accuracy
Logistic Regression	83.33%
Support Vector Machine	83.33%
Decision Tree	83.33%
K-Nearest Neighbors	83.33%





Conclusion

1. Success is highly dependent on **Orbit type** and **Payload mass**.
2. The **Decision Tree** model is recommended for its interpretability.
3. Reusing the first stage provides a massive competitive advantage by reducing launch costs by nearly 50%.